Status	Finished
Started	Tuesday, 20 October 2026, 6:13 PM
Completed	Tuesday, 20 October 2026, 6:18 PM
Duration	4 mins 36 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It is less efficient than custom sorting algorithms
- b. It sorts data out of the box efficiently
- c. It only works with integer arrays
- d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **2**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is searched for a specific element
- b. The parts are merged without sorting
- c. Each part is sorted independently
- d. The parts are ignored

The correct answer is: Each part is sorted independently

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Heap Sort
b. Bubble Sort
c. Quick Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Merging two datasets
b. Finding a specific element in a list
c. Arranging data in a specific order
d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **5**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It only works on small datasets
b. It simplifies the sorting process by using only one pass
c. It avoids the need for recursion
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **6**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Sorting data in a non-sequential manner
- b. Being the most efficient sorting algorithm
- c. Using the divide-and-conquer approach
- d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **7**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Decreases the efficiency of algorithms
- b. Provides a basis for other algorithms to work efficiently
- c. Makes code execution slower
- d. Makes data harder to manage

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Heap sort
- b. Merge sort
- c. Selection sort
- d. Quick sort

The correct answer is:
Merge sort

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Merge Sort
b. Bubble Sort
c. Binary Search
d. Linear Search

The correct answer is: Bubble Sort

Question **10**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

a. Bubbling up the smallest element
b. Sorting the entire list in one pass
c. Dividing the list into halves
d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **11**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By breaking the input into smaller parts and merging them
b. By comparing elements sequentially
c. By sorting data in a single pass
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Insertion Sort
- b. Bubble Sort
- c. Quick Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **13**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting is rarely used in programming
- b. Sorting helps in finding duplicates quickly
- c. Sorting makes it harder to search for items
- d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **14**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It bubbles up the largest element in each pass
- b. It recursively sorts subproblems
- c. It divides the list into sublists
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **15**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Sorting data sequentially
b. Dividing the input into parts, solving each part, and combining the solutions
c. Rearranging data without sorting
d. Sorting data in a single pass

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions